CSE 6730 Final Report: Comprehensive PDE Numerical Methods Comparison for Option Pricing

Repository Link

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Abstract

This project develops a unified framework for comparing numerical solution methods for option pricing PDEs across multiple financial models, validated through calibration to real market data. We implement and compare three classes of numerical methods: (1) explicit, implicit, and Crank-Nicolson finite difference schemes for the 1D Black-Scholes PDE, (2) four Alternating Direction Implicit schemes (Douglas, Craig-Sneyd, Modified Craig-Sneyd, and Hundsdorfer-Verwer) for the 2D Heston stochastic volatility PDE, and (3) IMEX schemes for jump-diffusion PIDEs. Each method is rigorously evaluated for stability, convergence, and computational efficiency. The project applies Levenberg-Marquardt optimization to calibrate Heston model parameters to OptionMetrics implied volatility surfaces, demonstrating how discretization choices affect both numerical accuracy and empirical model fit. Our implementation in Python provides a modular, well-documented codebase that serves as both a pedagogical tool for understanding PDE numerical methods and a practical framework for quantitative finance applications.

1 Project Description

In the world of quantitative finance, partial differential equations have been applied extensively to options pricing, with various numerical methods being introduced to handle certain challenges unique to specific stochastic options pricing models. The Black-Scholes model, despite its elegant closed-form solution, relies on the unrealistic assumption of constant volatility. More sophisticated models like Heston's stochastic volatility framework and Merton's jump-diffusion model address these limitations but introduce computational challenges that require specialized numerical techniques. From one-dimensional diffusion equations to two-dimensional PDEs with mixed derivatives and partial integro-differential equations with nonlocal jump operators, different models require increasingly sophisticated numerical approaches rather than a single generalizable solution method.

Our contribution is a unified framework that implements and rigorously compares numerical methods across three important option pricing models, validated through calibration to real market data from OptionMetrics. Specifically, we implement three classes of finite difference schemes for the one-dimensional Black-Scholes PDE, four alternating direction implicit (ADI) schemes for the two-dimensional Heston PDE, and implicit-explicit (IMEX) schemes for jump-diffusion partial integro-differential equations. Each method is evaluated not only on theoretical metrics such as convergence order and stability, but also on practical considerations including computational efficiency and calibration quality when fitted to market-observed implied volatility surfaces.

This project is fundamentally a simulation study focused on numerical methods for partial differential equations. While we utilize real market data, our primary objective is not statistical analysis but rather the systematic comparison of PDE solution techniques. By implementing these methods in a modular Python framework and validating them against both analytical solutions and empirical data, we demonstrate the interplay between numerical accuracy and empirical model fit.

2 Literature Review

A Heston stochastic model is a method to price options where both the price of the underlying stock and its volatility evolve randomly through time. It is a difficult model to calibrate as it is framed over 5 variables (dimensions) which are not entirely independent. The calibration approach proposed by the authors is a derived form of analytical gradient and natural optimization. They frame the problem as a nonlinear least-squares problem (comparing modeled price with actual price) by deriving a numerically continuous representation of the Heston characteristic function which avoids discontinuities, and formulating an analytical gradient which combined with Levenberg-Marquard (LM) algorithms enables fast gradient-based optimization. This issue is important because in option pricing time is a determinant factor for success as well as accuracy: the Heston model calibration landscape is irregular (proposed solutions don't seem to converge rapidly).

In 't Hout and Foulon (2010) analyze numerical methods for solving the two-dimensional Heston partial differential equation, which arises when modeling stochastic volatility with nonzero correlation between the asset price and its variance. Unlike the Black Scholes equation, the Heston PDE includes a mixed spatial derivative term that complicates stability and efficiency of standard finite difference methods. After discretizing space on nonuniform grids concentrated near the strike and low variance region, the PDE becomes a large stiff system of ordinary differential equations. The authors study several Alternating Direction Implicit schemes that split the operator into directional components, reducing computational complexity while maintaining accuracy. Through stability and convergence experiments, they show that appropriately designed ADI schemes achieve unconditional stability and second order convergence with com-

putational cost proportional to the number of grid points. This paper is directly relevant to our project because it provides a practical and theoretically grounded framework for implementing efficient PDE solvers for the Heston model. While the authors focus on numerical stability and convergence of specific schemes, we incorporate these methods into a broader comparative framework that evaluates performance across models and links numerical accuracy to empirical calibration outcomes.

Cont and Voltchkova (2005) present a finite difference method for solving parabolic partial integro-differential equations (PIDEs) that arise in option pricing when asset dynamics follow Lévy processes or jump-diffusion models. Unlike pure diffusion models, Lévy processes incorporate discontinuous jumps in asset prices, offering greater flexibility in capturing market phenomena such as volatility smiles and heavy-tailed return distributions. The authors propose an explicit-implicit (IMEX) time-stepping scheme that treats the differential component implicitly via standard finite difference discretization while handling the nonlocal integral term explicitly, thereby avoiding inversion of dense matrices. They establish localization error bounds for truncating the spatial domain and integral kernel, demonstrating exponential decay under integrability conditions on the Lévy measure. The paper rigorously analyzes the scheme's consistency, stability, and convergence using the viscosity solution framework, proving unconditional stability and monotonicity properties that ensure discrete comparison principles and arbitrage-free approximations. Convergence rates are derived under additional smoothness assumptions, and numerical experiments validate the method's accuracy and efficiency for both the Merton and variance Gamma models, including European and barrier options with smooth and nonsmooth payoffs.

3 Conceptual Model

This project implements and compares numerical solution methods for option pricing across three fundamental financial models: the Black-Scholes model for constant volatility, the Heston model for stochastic volatility, and the Merton jump-diffusion model for price discontinuities.

3.1 Black-Scholes Model (1D)

The Black-Scholes model assumes the underlying asset price S follows geometric Brownian motion with constant volatility σ:

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t \tag{1}$$

where r is the risk-free rate and W_t is a Wiener process. The option value $V(S,t)$ satisfies the parabolic PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0 \tag{2}$$

The Black-Scholes model serves as our baseline. While it has a closed-form solution for European options, solving the PDE numerically provides a foundation for understanding finite difference methods and validating our implementation before tackling more complex models. The one-dimensional spatial domain and absence of mixed derivatives make it an ideal testbed for explicit, implicit, and Crank-Nicolson schemes.

3.2 Heston Model (2D)

The Heston model introduces stochastic volatility to capture the empirically observed volatility smile and skew in option prices. The asset price S and variance v evolve according to:

$$dS_t = rS_t\,dt + \sqrt{v_t}S_t\,dW_t^S \tag{3}$$

$$dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v \tag{4}$$

where κ is the mean reversion speed, θ is the long-run variance, ξ is the volatility of volatility, and $dW_t^S dW_t^v = \rho\,dt$ with correlation ρ. The option value $V(S, v, t)$ satisfies the two-dimensional PDE:

$$\begin{aligned}\frac{\partial V}{\partial t} &= \frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2} + \rho\xi vS\frac{\partial^2 V}{\partial S\partial v} + \frac{1}{2}\xi^2 v\frac{\partial^2 V}{\partial v^2} \\ &\quad + rS\frac{\partial V}{\partial S} + \kappa(\theta - v)\frac{\partial V}{\partial v} - rV\end{aligned} \tag{5}$$

The key computational challenge is the mixed derivative term $\partial^2 V/\partial S\partial v$, which couples the two spatial directions. This term arises from the correlation ρ between asset returns and volatility changes. Standard implicit finite difference methods would require solving a dense $(N_S \times N_v) \times (N_S \times N_v)$ linear system at each time step, which is computationally prohibitive. Alternating Direction Implicit (ADI) schemes circumvent this by splitting the 2D problem into a sequence of 1D problems.

3.3 Merton Jump-Diffusion Model (1D PIDE)

The Merton model extends Black-Scholes by adding discrete jumps to the asset price dynamics:

$$dS_t = (r - \lambda m)S_t\,dt + \sigma S_t\,dW_t + S_t dJ_t \tag{6}$$

where J_t is a compound Poisson process with intensity λ and log-normal jump sizes. The option value satisfies a partial integro-differential equation (PIDE):

$$\begin{aligned}\frac{\partial V}{\partial t} &= \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + (r - \lambda m)S\frac{\partial V}{\partial S} - (r + \lambda)V \\ &\quad + \lambda\int_{-\infty}^{\infty} V(Se^y, t)\nu(y)\,dy\end{aligned} \tag{7}$$

where $\nu(y)$ is the jump size density and $m = E[e^Y - 1]$ is the expected relative jump size.

The integral term represents the expected value contribution from jumps and creates a nonlocal coupling across the spatial grid. This term cannot be treated implicitly without constructing and inverting a dense matrix. The IMEX (Implicit-Explicit) approach treats the diffusion operator implicitly for stability while handling the integral operator explicitly to avoid dense linear systems.

4 Simulation

Each of the financial models above presents distinct computational challenges that require specialized numerical techniques.

4.1 Finite Difference Schemes for Black-Scholes

We implement three classical finite difference schemes for the one-dimensional Black-Scholes PDE:

- **Explicit scheme**: Forward Euler in time, central differences in space. Simple to implement but conditionally stable, requiring $\Delta t \leq \Delta S^2/(\sigma^2 S_{\max}^2)$.
- **Implicit scheme**: Backward Euler in time, central differences in space. Unconditionally stable but only first-order accurate in time. Requires solving a tridiagonal system at each time step.
- **Crank-Nicolson scheme**: Averages explicit and implicit schemes ($\theta = 0.5$). Second-order accurate in both time and space while maintaining unconditional stability. Also requires tridiagonal solves.

All three schemes discretize the spatial second derivative using central differences:

$$\frac{\partial^2 V}{\partial S^2} \approx \frac{V_{i+1} - 2V_i + V_{i-1}}{\Delta S^2} \tag{8}$$

This produces a tridiagonal system structure that can be solved efficiently using the Thomas algorithm in $O(N)$ operations.

4.2 ADI Schemes for Heston

Alternating Direction Implicit (ADI) methods decompose the two-dimensional problem into a sequence of one-dimensional sweeps. At each time step, the method alternates between treating the S-direction implicitly while keeping the v-direction explicit, and vice versa. We implement four ADI schemes from In 't Hout and Foulon [3]:

- **Douglas scheme**: The simplest ADI approach with two sweeps per time step. First-order accurate in time, $O(\Delta t, \Delta S^2, \Delta v^2)$. The mixed derivative term is treated explicitly using values from the previous time step, which can cause instability for large correlation $|\rho|$.

- **Craig-Sneyd scheme**: Adds two correction sweeps to achieve second-order temporal accuracy, $O(\Delta t^2, \Delta S^2, \Delta v^2)$. The correction steps update the mixed derivative using intermediate values, significantly improving stability for strong correlation.
- **Modified Craig-Sneyd (MCS) scheme**: Modifies the treatment of the mixed derivative by applying half in the first sweep and half in the second sweep, rather than applying all of it initially. This more balanced approach enhances stability, particularly for large time steps.
- **Hundsdorfer-Verwer (HV) scheme**: Uses optimally weighted splitting of operators with parameters γ, α, and β chosen to minimize error constants. Achieves the best accuracy among ADI schemes at the cost of slightly increased complexity.

Each directional sweep requires solving N tridiagonal systems of size M, yielding $O(NM)$ complexity per sweep compared to $O((NM)^3)$ for a full implicit 2D scheme. The key algorithmic difference between schemes lies in how they handle the mixed derivative term $\rho\xi v S \partial^2 V/\partial S \partial v$.

4.3 IMEX Schemes for Jump-Diffusion

The Implicit-Explicit (IMEX) time-stepping method splits the PIDE operator into local (differential) and nonlocal (integral) components:

$$\frac{\partial V}{\partial t} = \mathcal{L}_{\text{diff}}(V) + \mathcal{I}_{\text{jump}}(V) \tag{9}$$

where $\mathcal{L}_{\text{diff}}$ contains the diffusion and convection terms, and $\mathcal{I}_{\text{jump}}$ is the integral operator.

The IMEX scheme as developed by Cont and Voltchkova [1] treats the diffusion operator implicitly for unconditional stability while evaluating the jump integral explicitly:

$$(I - \Delta t \mathcal{L}_{\text{diff}})V^{n+1} = V^n + \Delta t \mathcal{I}_{\text{jump}}(V^n) \tag{10}$$

The integral term is computed via numerical quadrature (typically Gaussian quadrature or adaptive Simpson's rule) over the jump size distribution. This approach avoids the dense linear systems that would arise from implicit treatment of the integral while maintaining the stability benefits of implicit time-stepping for the diffusion part.

4.4 Monte Carlo Methods

Under the Black-Scholes framework, the asset price follows geometric Brownian motion discretized via the exact log-normal solution $S_{t+\Delta t} = S_t \exp\Big((\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\, Z\Big)$, $Z \sim \mathcal{N}(0,1)$, with option prices estimated as $\hat{V} = e^{-rT}\frac{1}{M}\sum_{i=1}^{M} \max(S_T^{(i)} - K, 0)$

over M paths. The Heston model introduces stochastic variance $dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v$ correlated with the asset process at coefficient ρ, with correlated increments constructed from independent draws $Z_1, Z_2 \sim \mathcal{N}(0,1)$ and the full truncation scheme $\max(v_t, 0)$ applied to prevent negative variance. The Merton jump-diffusion model further augments this with a compound Poisson process at rate λ, where log-jumps $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$ are accumulated per step and the drift compensator $\bar{k} = e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1$ preserves the martingale property under the risk-neutral measure.

4.5 Calibration Framework

Model parameters are estimated by minimizing the sum of squared differences between model-implied and market-observed option prices. For the Heston model, we calibrate the five parameters $\{\kappa, \theta, \xi, \rho, v_0\}$ to match the implied volatility surface. The objective function is:

$$\min_{\theta} \sum_{i=1}^{N} w_i \left(V_{\text{market}}^i - V_{\text{model}}^i(\theta)\right)^2 \tag{11}$$

where w_i are weights (typically inverse of bid-ask spread or vega) and the sum is over N liquid options across various strikes and maturities.

We employ the Levenberg-Marquardt algorithm, which combines gradient descent with Gauss-Newton updates. Following Cui et al. [2], we compute gradients of option prices with respect to model parameters numerically via finite differences on the PDE solution. This approach is significantly faster than recomputing the entire PDE solution for each parameter perturbation.

5 Experimental Results and Validation

5.1 Experimental Procedure

Our experimental procedure was organized as follows.

1. Download and clean OptionMetrics data for SPX, RUT, TSLA, and PLTR over a multi-week window, retaining contracts expiring between 30-60 days

2. Filter contracts by liquidity and basic quality controls, including bid–ask spread, volume, implied volatility range, and log-moneyness bounds, with per-ticker thresholds (stricter for the indices, looser for TSLA and PLTR).

3. Calibrate Heston parameters for each underlying using a stratified sample of up to 75 cleaned contracts

4. Calibrate Merton jump-diffusion parameters for each underlying using an exhaustive grid search over $(\lambda, \mu_J, \sigma_J)$; each grid point is evaluated by pricing the sample contracts with the IMEX–Euler PIDE solver, and the parameter set minimizing mean absolute error is selected

5. Price up to 75 contracts per ticker under calibrated parameters using all numerical methods (including Monte Carlo methods) in a single parallel queue, saving results to csv format
6. Aggregate per-contract pricing results into a summary CSV and compute model prices versus market mid-prices
7. Compute model Greeks (delta, gamma, etc.) for each underlying and compare against the OptionMetrics reference values
8. Run synthetic convergence and stress studies on a benchmark contract ($S = K = 100$, $T = 1$ year) with no market data, using closed-form Black–Scholes prices, Heston characteristic-function prices, and a fine-grid IMEX solution as references for each model class

To improve comparability across schemes, all methods within a model class used the same contract inputs and broadly similar domain construction rules. For Black-Scholes and Merton, the stock-price grid bounds were chosen from a lognormal heuristic centered around the observed spot with width proportional to $\sigma\sqrt{T}$. For Heston, the stock-price and variance domains were chosen using the current spot, strike, initial variance, and long-run variance.

5.2 Studies Performed

5.2.1 Market-Data Pricing Studies

European options on SPX, RUT, TSLA, and PLTR were priced using each model's numerical solver against a cleaned OptionMetrics sample filtered for liquidity and quality. Black-Scholes experiments employed explicit, implicit, and Crank-Nicolson finite difference schemes; Heston experiments used four ADI schemes (Douglas, Craig-Sneyd, Modified Craig-Sneyd, and Hundsdorfer-Verwer); and Merton experiments used an IMEX scheme for the jump-diffusion PIDE. For each contract, we recorded the model price, absolute error, and squared error relative to the market mid price, with Monte Carlo pricers run in parallel as an independent cross-check against the PDE and PIDE results.

5.2.2 Calibration Studies

We performed two model calibration studies across all four underlyings. For the Heston model, we calibrated the parameter vector

$$\theta_{\text{Heston}} = (\kappa,\, \theta,\, \xi,\, \rho,\, v_0)$$

by minimizing weighted squared pricing errors over a filtered set of market options, with greater weight placed on lower-priced contracts through normalized inverse-price weighting. Calibration was performed using nonlinear least squares, after which the calibrated parameters were validated by repricing the contracts with both the semi-analytic characteristic-function formula and the PDE solver. Table 1 reports the calibrated parameters for each underlying.

Table 1: Calibrated Heston parameters and repricing errors by underlying. The Feller condition $2\kappa\theta > \xi^2$ is evaluated for each set.

Ticker	κ	θ	ξ	ρ	v_0	Feller?	CF MAE	PDE MAE
SPX	2.186	0.155	0.824	-0.902	0.030	Marginal	22.358	42.379
RUT	14.989	0.001	0.173	-0.003	0.137	Satisfied	16.031	14.861
TSLA	15.000	0.500	1.000	-0.059	0.285	Satisfied	4.021	4.706
PLTR	14.845	0.500	0.010	≈ 0	0.500	Satisfied	7.612	7.393

The SPX calibration yields $2\kappa\theta \approx \xi^2 \approx 0.678$, placing the solution at the boundary of the Feller condition $2\kappa\theta > \xi^2$, which guarantees strict positivity of the variance process v_t. When this condition is marginal, v_t can touch zero and the full truncation scheme introduces a discretization bias at each such event, explaining the elevated SPX PDE MAE (42.379) relative to the characteristic-function MAE (22.358); the semi-analytic pricer is unaffected by v_t discretization, whereas the PDE solver accumulates bias whenever variance touches zero. The remaining underlyings satisfy the Feller condition comfortably and show tighter CF-to-PDE agreement, and the synthetic convergence experiments in Section 5.2.3 use parameters well within the condition, confirming that the elevated SPX error is a calibration artifact rather than a defect in the numerical method.

For the Merton jump-diffusion model, we performed a grid search over

$$\theta_{\text{Merton}} = (\lambda,\, \mu_J,\, \sigma_J),$$

where λ is the jump intensity and (μ_J, σ_J) describe the log-jump distribution. Each candidate parameter set was evaluated on a calibration subsample of 60 contracts per underlying, and the best set was selected by mean absolute error. Table 2 reports the calibrated parameters.

Table 2: Best-fit Merton jump-diffusion parameters by underlying, selected by minimum MAE over the calibration grid.

Ticker	λ	μ_J	σ_J	MAE	RMSE
SPX	0.000	0.000	0.010	3.035	4.220
RUT	0.010	0.000	0.010	1.399	1.973
TSLA	0.010	-0.020	0.050	0.329	0.472
PLTR	0.000	0.000	0.010	0.172	0.216

The SPX and PLTR calibrations selected $\lambda = 0$, meaning the grid search found no benefit from adding jumps to the diffusion for those underlyings on this sample. For TSLA, the best-fit parameters include a nonzero intensity and a negative mean jump, which is consistent with the well-documented left-skew in single-stock implied volatility surfaces. The narrow spread of MAE values across the PLTR grid reflects the fact that PLTR options in the calibration sample were short-dated and relatively inexpensive, leaving little room for jump parameters to differentiate themselves.

5.2.3 Convergence and Stability Studies

To isolate numerical behavior from market noise, we ran synthetic benchmark studies on a single at-the-money European call with $S_0 = K = 100$ and $T = 1$. For Black-Scholes, the reference price was the closed-form analytical solution (10.4506). For Heston, the reference price was the semi-analytic characteristic-function price (10.4033). For Merton, the reference price was a fine-grid IMEX solution (10.2959). We varied spatial and temporal resolution across four refinement levels and recorded price, runtime, absolute error, and observed convergence order. Stress scenarios were also run with $\rho = -0.90$ and $\xi = 0.60$ for Heston and $\lambda = 1.00$ for Merton. For visualizations highlighting the convergence studies, see section B in Appendix.

5.2.4 Monte Carlo Studies

Monte Carlo pricers were implemented for Black-Scholes, Heston, and Merton across all four underlyings. These experiments served two purposes: providing an independent pricing methodology based directly on the underlying SDEs rather than PDE discretization, and quantifying sampling uncertainty through standard errors and 95% confidence intervals. The Black-Scholes and Merton Monte Carlo runs used antithetic variates; the Heston runs used correlated Brownian shocks with the Euler-Maruyama discretization.

5.3 Validation Strategy

We validated the simulator in three ways.

5.3.1 Validation Against Analytical or Semi-Analytical References

For Black-Scholes, validation was straightforward because the exact European option price is known in closed form. For Heston, we used the semi-analytic characteristic-function price as the primary validation reference. For Merton, we validated the IMEX solver against a high-resolution fine-grid numerical solution.

5.3.2 Cross-Method Validation

We also compared different numerical methods applied to the same contracts. For Black-Scholes, the three PDE schemes agree closely across all underlyings, and the Monte Carlo estimates are statistically consistent with the PDE prices once sampling error is accounted for. For Heston, the four ADI schemes produce comparable errors on a given underlying, and the Monte Carlo Heston prices track the PDE results. The Merton IMEX solver and Merton Monte Carlo likewise produce similar MAE values on each underlying.

5.3.3 Empirical Validation on Market Data

The most practically relevant validation came from comparing model outputs with observed market mid prices across four underlyings. Because the models

are stylized approximations, this is not an exact mathematical validation, but it allows us to assess whether each model is calibrated to a reasonable regime and whether errors are systematic.

5.4 Output Analysis

At the contract level, each pricing run produced the model price, absolute pricing error, squared pricing error, and, for Monte Carlo methods, a standard error and 95% confidence interval. At the aggregate level, we summarized each experiment using mean absolute error (MAE) and root mean squared error (RMSE). For convergence studies, we additionally recorded runtime and observed convergence order, computed as

$$p_{\text{obs}} = \frac{\log(e_{h_1}/e_{h_2})}{\log(h_1/h_2)},$$

where e_{h_1} and e_{h_2} are the absolute errors at two successive grid resolutions. For Monte Carlo, the option price estimator is the discounted sample mean of terminal payoffs,

$$\hat{V} = e^{-rT}\frac{1}{M}\sum_{i=1}^{M}\Pi_i,$$

with standard error $\text{SE}(\hat{V}) = s_\Pi e^{-rT}/\sqrt{M}$ and 95% confidence interval $\hat{V} \pm 1.96 \cdot \text{SE}(\hat{V})$.

5.5 Main Quantitative Findings

5.5.1 Black-Scholes Pricing

The Black-Scholes finite-difference schemes were numerically consistent across all three methods and all four underlyings. Table 3 summarizes the results. For SPX, MAE ranged from 1.883 (explicit) to 1.896 (implicit), with RMSE near 2.84 for all three. For RUT, MAE ranged from 1.119 to 1.122 with RMSE near 1.49. For TSLA, errors were substantially smaller (MAE $\approx$ 0.25–0.27, RMSE $\approx$ 0.30–0.35), reflecting the lower absolute price level of TSLA options in the sample. For PLTR, errors were smallest of all (MAE $\approx$ 0.110, RMSE $\approx$ 0.133). The three schemes agree to within rounding on each underlying, which is consistent with all three solving the same one-dimensional PDE and differing only in stability and truncation properties. Monte Carlo Black-Scholes produced lower MAE than the PDE schemes on each underlying: 0.746 (SPX), 0.826 (RUT), 0.148 (TSLA), and 0.076 (PLTR).

5.5.2 Heston Pricing

The Heston results are dominated by the calibration quality of the underlying-specific parameters, particularly the proximity of the calibrated parameters to

Table 3: Black-Scholes pricing results by underlying and scheme.

Ticker	Scheme	Contracts	MAE	RMSE
SPX	Explicit	75	1.883	2.838
SPX	Implicit	75	1.896	2.837
SPX	Crank-Nicolson	75	1.885	2.838
SPX	Monte Carlo	75	0.746	0.989
RUT	Explicit	75	1.119	1.491
RUT	Implicit	75	1.122	1.492
RUT	Crank-Nicolson	75	1.120	1.491
RUT	Monte Carlo	75	0.826	1.239
TSLA	Explicit	73	0.244	0.303
TSLA	Implicit	75	0.266	0.345
TSLA	Crank-Nicolson	75	0.265	0.344
TSLA	Monte Carlo	75	0.148	0.183
PLTR	Explicit	75	0.110	0.133
PLTR	Implicit	75	0.110	0.134
PLTR	Crank-Nicolson	75	0.110	0.133
PLTR	Monte Carlo	75	0.076	0.084

the Feller boundary. Table 4 reports the PDE and Monte Carlo MAE and RMSE by underlying and scheme.

The large SPX Heston errors (MAE $\approx$ 19–21) reflect the marginal Feller condition discussed in Section 5.2.3: when $2\kappa\theta \approx \xi^2$, the variance process frequently approaches zero, the full truncation introduces repeated discretization bias, and the net effect is substantially elevated PDE error relative to what the semi-analytic pricer yields under the same parameters. The RUT, TSLA, and PLTR calibrations clearly satisfy the Feller condition, and the corresponding Heston errors are markedly lower. Within each underlying, the four ADI schemes produce very similar MAE and RMSE values, with differences at most a few tenths of an MAE unit, confirming that the choice of ADI scheme is secondary to parameter quality on these market-data experiments.

5.5.3 Merton Pricing

Table 5 summarizes the Merton IMEX PIDE and Monte Carlo results. The Merton IMEX solver produced MAE of 3.063 and RMSE of 4.211 on SPX, 1.587 and 2.153 on RUT, 0.321 and 0.447 on TSLA, and 0.173 and 0.214 on PLTR. The Merton Monte Carlo results were close to the Black-Scholes Monte Carlo results on each underlying, which is consistent with the calibrated jump parameters being small or zero for SPX and PLTR.

5.5.4 Convergence and Stability Studies

Tables 6 and 7 summarize the synthetic benchmark Heston ADI comparison in terms of final absolute error, runtime, and qualitative assessment, split by

Table 4: Heston pricing results by underlying and scheme.

Ticker	Scheme	Contracts	MAE	RMSE
SPX	Douglas	75	20.617	25.862
SPX	Craig-Sneyd	75	20.625	25.891
SPX	Modified Craig-Sneyd	75	21.306	26.496
SPX	Hundsdorfer-Verwer	75	19.463	25.167
SPX	Monte Carlo	75	24.029	30.293
RUT	Douglas	75	7.363	9.305
RUT	Craig-Sneyd	75	7.363	9.305
RUT	Modified Craig-Sneyd	75	7.430	9.365
RUT	Hundsdorfer-Verwer	75	7.248	9.220
RUT	Monte Carlo	75	7.490	9.408
TSLA	Douglas	75	0.860	0.982
TSLA	Craig-Sneyd	75	0.860	0.981
TSLA	Modified Craig-Sneyd	75	0.859	0.980
TSLA	Hundsdorfer-Verwer	75	0.862	0.984
TSLA	Monte Carlo	75	1.028	1.106
PLTR	Douglas	75	1.989	2.062
PLTR	Craig-Sneyd	75	1.989	2.062
PLTR	Modified Craig-Sneyd	75	1.992	2.065
PLTR	Hundsdorfer-Verwer	75	1.983	2.056
PLTR	Monte Carlo	75	1.899	1.969

refinement study and stress scenario respectively.

Across both spatial and temporal refinement, the Modified Craig-Sneyd scheme consistently produced the smallest final absolute error, while Douglas was the fastest. Craig-Sneyd offered an intermediate tradeoff between accuracy and runtime. Hundsdorfer-Verwer was the least accurate of the four on both refinement dimensions despite comparable or higher runtimes, though it remained numerically stable throughout. Under the stress scenario ($\rho = -0.90$, $\xi = 0.60$), all four schemes remained stable but showed elevated errors, with Modified Craig-Sneyd again the most accurate and Hundsdorfer-Verwer the least.

For Black-Scholes, the spatial refinement experiments showed overall improvement for the implicit and Crank-Nicolson schemes, though convergence was non-monotone; the explicit scheme showed no clear convergence under standalone temporal refinement and required stable coupled space-time refinement instead. For Merton, the IMEX Euler scheme showed overall improvement under spatial refinement (final absolute error 0.0545 against reference 10.2959), no clear monotone convergence under standalone temporal refinement, and remained stable under the high-jump stress test ($\lambda = 1.00$), producing a final absolute error of 0.213 against reference 12.295.

Table 5: Merton jump-diffusion pricing results by underlying.

Ticker	Scheme	Contracts	MAE	RMSE
SPX	IMEX Euler	75	3.063	4.211
SPX	Monte Carlo	75	0.746	0.989
RUT	IMEX Euler	75	1.587	2.153
RUT	Monte Carlo	75	0.827	1.239
TSLA	IMEX Euler	75	0.321	0.447
TSLA	Monte Carlo	75	0.150	0.186
PLTR	IMEX Euler	75	0.173	0.214
PLTR	Monte Carlo	75	0.076	0.084

Table 6: Heston ADI scheme comparison: spatial and temporal refinement ($S_0 = K = 100$, $T = 1$, reference price = 10.4033).

Scheme	Scenario	Final Abs. Error	Final Runtime (s)	Assessment
Douglas	Spatial	0.8315	2.8669	Clear convergence
Craig-Sneyd	Spatial	0.8268	5.1725	Clear convergence
Modified Craig-Sneyd	Spatial	0.5759	6.0602	Clear convergence
Hundsdorfer-Verwer	Spatial	1.2618	5.7016	Clear convergence
Douglas	Temporal	0.7886	15.5314	Clear convergence
Craig-Sneyd	Temporal	0.7850	28.3247	Clear convergence
Modified Craig-Sneyd	Temporal	0.5448	33.7469	Clear convergence
Hundsdorfer-Verwer	Temporal	1.2012	31.1534	Clear convergence

5.6 Interpretation of Inputs

Our input modeling choices were intended to balance realism and tractability. Covering four underlyings with different volatility regimes (SPX as a broad index, RUT as a small-cap index, TSLA as a high-volatility single stock, and PLTR as a lower-priced growth stock) allowed us to stress-test the calibration pipeline across meaningfully different market conditions. We used market-implied volatility as the diffusion input for Black-Scholes and Merton pricing, while the Heston and Merton structural parameters were estimated through calibration. This design reflects the role of each model in practice: Black-Scholes is used as a local benchmark conditional on market volatility, while Heston and jump-diffusion models are useful precisely because their additional parameters can be fit to observed cross-sections and then used to price instruments outside the calibration sample.

Overall, the experimental framework allowed us to evaluate the simulator from three angles at once: mathematical correctness against known benchmarks, numerical quality through convergence and stability studies, and practical relevance through calibration and pricing on market data across multiple underlyings.

Table 7: Heston ADI scheme comparison: stress scenario ($\rho = -0.90$, $\xi = 0.60$; $S_0 = K = 100$, $T = 1$, reference price = 8.3697).

Scheme	Scenario	Final Abs. Error	Final Runtime (s)	Assessment
Douglas	Stress	1.0742	1.8987	Stable, reduced accuracy
Craig-Sneyd	Stress	1.0722	3.4360	Stable, reduced accuracy
Modified Craig-Sneyd	Stress	0.9264	4.0698	Stable, reduced accuracy
Hundsdorfer-Verwer	Stress	1.3229	3.7635	Stable, reduced accuracy

6 Discussion and Conclusions

Across all underlyings and model classes, the dominant finding was that calibration quality mattered far more than the choice of numerical scheme: competing PDE and PIDE schemes produced nearly identical pricing errors within each model class, while errors varied substantially across underlyings based on how well the calibrated parameters matched the implied volatility surface. The SPX Heston results illustrated this sharply, where the marginal Feller condition produced a large semi-analytic-to-PDE gap due to repeated full-truncation bias whenever the variance touched zero, while TSLA and PLTR, whose parameters clearly satisfied the Feller condition, showed tight agreement across all four ADI schemes. Monte Carlo pricing tracked PDE and PIDE results closely on well-conditioned underlyings, confirming internal consistency across both computational approaches. The Merton calibration revealed a subtler point: the grid search selected $\lambda = 0$ for SPX and PLTR, indicating the jump component added no explanatory power on these short-dated contracts, while TSLA's negative mean jump $\mu_J = -0.02$ is economically interpretable as downside crash risk. For future work, we recommend replacing the full truncation scheme with the Quadratic-Exponential discretization of Andersen (2008) for near-zero variance, substituting a proper nonlinear least squares calibration for the Merton grid search, extending the dataset to longer maturities and wider moneyness ranges, and adding control variates using the Black-Scholes price as a baseline to reduce Monte Carlo standard errors without increasing path count.

References

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[2] Yiran Cui, Sebastian del Baño Rollin, and Guido Germano. Full and fast calibration of the Heston stochastic volatility model. *European Journal of Operational Research*, 263(2):625–638, 2017.

[3] Karel J. in 't Hout and Sophine Foulon. ADI finite difference schemes for option pricing in the Heston model with correlation. *International Journal of Numerical Analysis and Modeling*, 7(2):303–320, 2010.

[2] [3] [1]

AI Statement: AI has been used so far in the project for the purpose of identifying ideas (specifically conducting feasibility analysis of ideas that the group had come up with) and to improve the readability of the literature review, checkpoints, and final report. While coding, AI tools such as Claude, ChatGPT, and Cursor were used to debug and add documentation.

A Division of Labor

Sarang was responsible for repository setup and data acquisition, including configuring the WRDS connection and building the OptionMetrics data pipeline. On the modeling side, he implemented the Heston model in full: the five-parameter calibration routine, all four ADI schemes (Douglas, Craig-Sneyd, Modified Craig-Sneyd, and Hundsdorfer-Verwer), and the Greeks computation for the Heston PDE. He also wrote the sections of the paper related to ADI schemes or the Heston model analysis. He also designed and implemented the parallel processing infrastructure and the main orchestrator script that coordinates model runs across underlyings and schemes.

Pascal was responsible for the Black-Scholes model, implementing the explicit, implicit, and Crank-Nicolson finite difference schemes along with the corresponding Monte Carlo pricer. He wrote the conceptual model section of the report, which develops the mathematical formulation of all three financial models. He also led the literature review, coordinated the presentation, and contributed helper functions used across the analysis pipeline.

Michael was responsible for the Merton jump-diffusion model, implementing the IMEX finite difference scheme for the PIDE, the grid-search calibration routine, and the corresponding Monte Carlo pricer. He led the convergence and stability analysis across all three models and implemented the Greeks computations for both Black-Scholes and the jump-diffusion model. He also wrote the report sections corresponding to the Merton model and the convergence studies.

B Convergence Plots

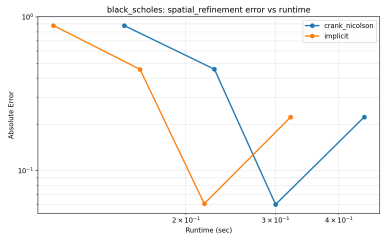

Figure 1: Black-Scholes Spatial Error vs Runtime

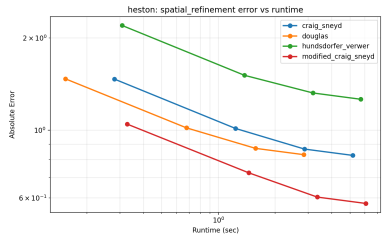

Figure 2: Heston Spatial Error vs Runtime

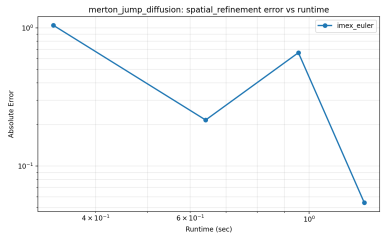

Figure 3: Merton Spatial Error vs Runtime

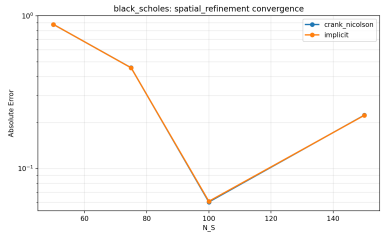

Figure 4: Black-Scholes Spatial Error vs Spatial Discretization

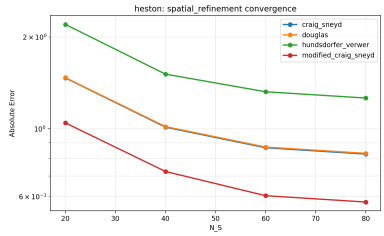

Figure 5: Heston Spatial Error vs Spatial Discretization

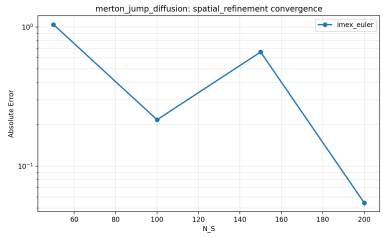

Figure 6: Merton Spatial Error vs Spatial Discretization

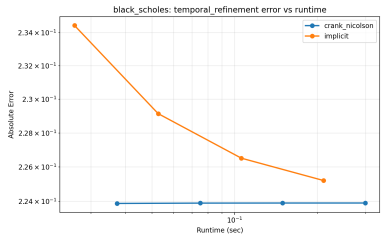

Figure 7: Black-Scholes Temporal Error vs Runtime

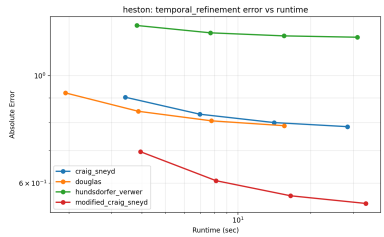

Figure 8: Heston Temporal Error vs Runtime

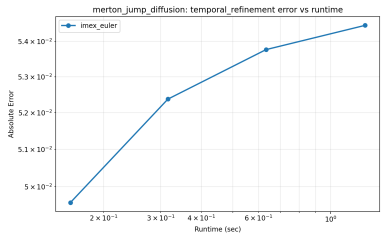

Figure 9: Merton Temporal Error vs Runtime

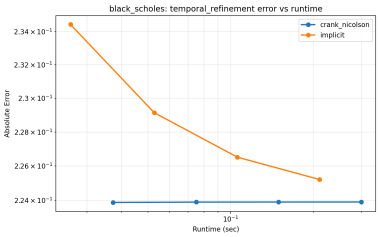

Figure 10: Black-Scholes Temporal Error vs Temporal Discretization

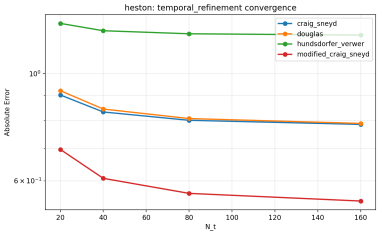

Figure 11: Heston Temporal Error vs Temporal Discretization

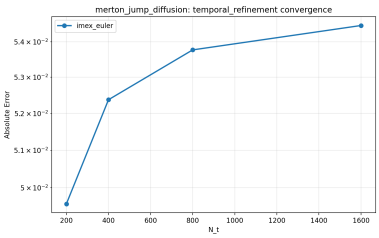

Figure 12: Merton Temporal Error vs Temporal Discretization